

Toronto Bicycle Theft

Part I	Part II	Part III	
Task 1	+ Task 4		
5	5		
Toronto bicycle theft reports	2014–2023	31,833	140
Basis OLS	Task 4	Basis Ridge	R ²

PPT “ ”

PPT			
1	0:00–1:10		
2	1:10–2:15		downtown neighbourhoods
3	2:15–4:05	Task 4	Ridge
4	4:05–5:00		

697	140	5	5
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- Task 1 Task 4
- Basis OLS Basis Ridge L2
- OLS RMSE Ridge MAE Ridge
- reported theft counts

Part I ·

English

PPT

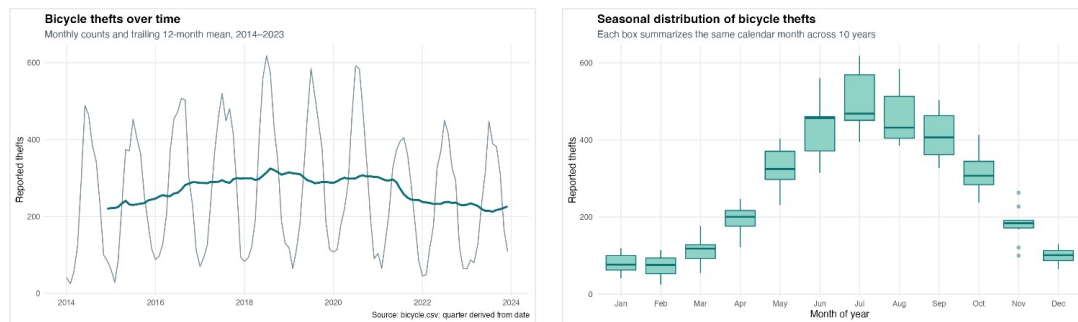
1 ·

| 0:00–1:10

Toronto thefts peak in summer and downtown

Name: _____ Student ID: _____

31,833 incidents · 140 areas · 2014–2023



TEMPORAL SIGNAL

Reported bicycle thefts show strong seasonality: July has the highest average count, while winter months remain below 100 incidents.

1

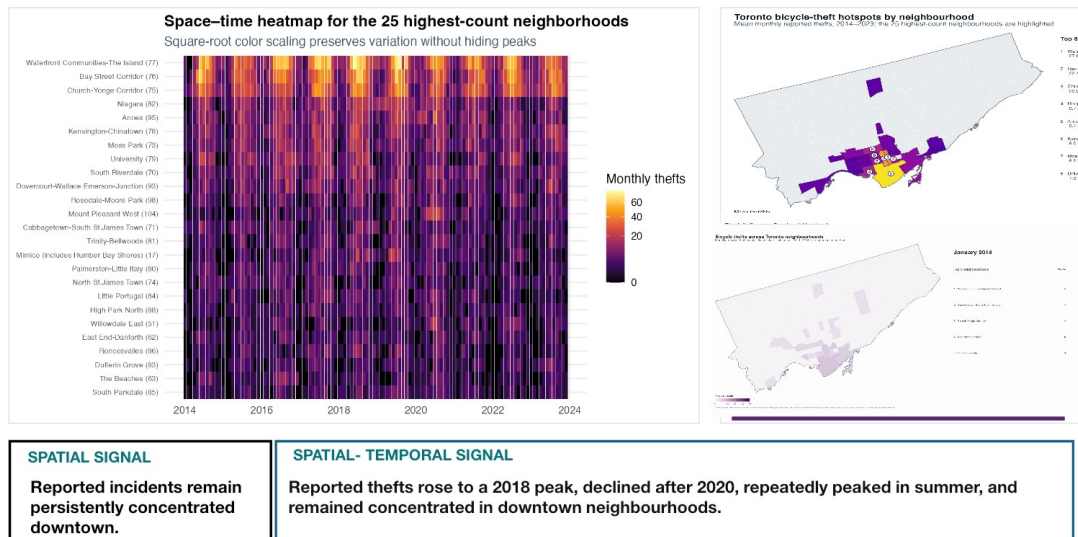
Good morning. My project studies reported bicycle thefts in Toronto from 2014 to 2023. The raw data contain 31,833 incidents across 140 neighbourhoods. I aggregated these events into a neighbourhood-by-month panel, where each row records the number of reported thefts in one neighbourhood during one month. This structure lets us study temporal change and spatial variation together. On the left, the citywide monthly series shows repeated sharp peaks, mostly during the warmer part of each year. However, the peak height is not constant across years, so the data contain both a seasonal cycle and longer-run change. The monthly distribution on the right confirms that the typical count rises from winter into summer and then falls toward the end of the year. The main message from this slide is that Toronto bicycle theft is seasonal, but the strength of the seasonal peak changes over time.

2 ·

| 1:10–2:15

Downtown Toronto remains the theft hotspot

Waterfront 27.6/month · Bay Street 22.4 · Church–Yonge 20.0 · Top three = 26.4% of all incidents



2

downtown

5

Slide two adds the spatial dimension and completes the required Task 1. In the space-time heatmap, each row is one high-count neighbourhood, each column is one month, and brighter colours indicate more reported thefts. Many rows brighten during similar months, showing that the seasonal peak is not an isolated event in only one area. At the same time, the brightest cells remain concentrated in a relatively small group of neighbourhoods. The map at the upper right returns these long-run averages to their geographic locations and shows that the main hotspots are concentrated around downtown Toronto. The short animation below shows how the pattern changes month by month from 2014 to 2023. Hotspot intensity rises and falls, but the broad spatial ordering remains persistent. Therefore, Task 1 provides two connected findings: a warm-season temporal peak and a stable downtown spatial concentration.

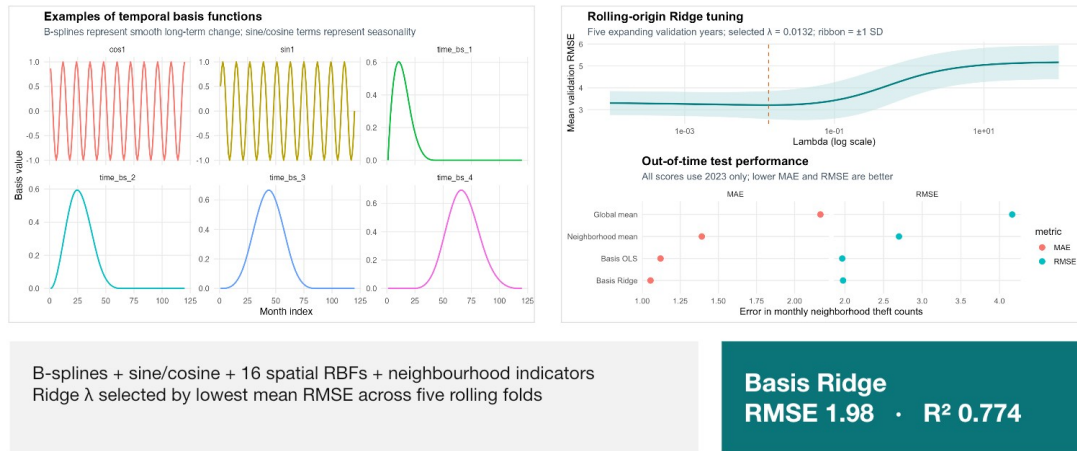
downtown neighbourhoods

3 · Task 4

| 2:15–4:05

Basis functions encode time and space

Count → log1p → temporal + seasonal + spatial bases → OLS / Ridge Rolling validation 2018–2022 → Final test 2023



3

B-spline

OLS Ridge

 R^2 “77.4%”

I then address Task 4 by building a linear regression model with basis functions. The response is the monthly theft count for each neighbourhood. Because a small number of neighbourhood-months have very high counts, I model log one plus the count. This reduces the influence of extreme peaks while preserving zero observations. The model is still linear in its coefficients, but the design matrix contains transformed features rather than only raw time and coordinates. First, cubic B-splines represent smooth long-run change using several local curves. Second, sine and cosine terms represent annual and half-year seasonal cycles. Third, sixteen radial basis functions, or RBFs, convert distance from selected spatial centres into smooth measures of proximity. I also retain neighbourhood indicators to capture persistent area differences. In other words, the basis functions convert nonlinear time, seasonal, and spatial patterns into columns of a design matrix, and the regression forms a linear combination of those columns. To tune Ridge without future information leakage, I use rolling-origin cross-validation with five expanding-window folds. The validation years are 2018 through 2022; for each fold, the model sees only earlier years. I average validation RMSE across the five folds and select lambda 0.0132. I then refit on all data from 2014 through 2022 and evaluate once on the untouched 2023 test set. Basis OLS is the standard basis-function linear regression required by Task 4. Basis Ridge uses the same design matrix but adds a squared-coefficient penalty, which stabilises coefficients when basis columns are correlated; it is not a separate black-box model. On the 2023 test set, Basis OLS has an RMSE of 1.97 and an R-squared of 0.776. Basis Ridge has an RMSE of 1.98 and an R-squared of 0.774, with the lower MAE of 1.05. Their predictive performance is almost identical, so I retain Ridge as the stable primary model while reporting OLS transparently.

Ridge

4 ·

| 4:05–5:00

Seasonality is captured, but peaks are missed



4

The final slide checks what the model actually learned. The observed and Ridge prediction curves share the main annual cycle. Ridge reduces RMSE by 26.8 percent relative to the neighbourhood-mean baseline, and both basis models clearly outperform the citywide-mean and neighbourhood-mean baselines. However, the largest summer peaks are still underestimated. The residual map also shows that most neighbourhood errors are modest, while extreme months and a small number of hotspots remain the main difficulty. Overall, the project completes Task 1 and Task 4: the visualisations identify warm-season peaks and persistent downtown concentration, while basis-function regression combines long-run, seasonal, and spatial signals in one interpretable predictive framework. Finally, these are associations in reported theft counts. They do not establish a causal effect of season or location on bicycle theft. Thank you.

Part II ·

“ → → → ”

1. neighbourhood-month

“ ”

$y_{it} =$				
i	t			
140	120	16,800		y_{it}

2.

“ ” “ ” “ ”

$\sum_i y_{it}$	$\sum_t (y_{it})$	y_{it}		
Slide 1	Slide 2	25	2014–2023	
Task 1	30%			

3. log(1+y)

$z = \log(1+y)$		$\hat{y} = \max(0, \exp(\hat{z})-1)$		
OLS	Ridge	log1p	MAE	RMSE R ²

4.

$$z = X\beta + \varepsilon \quad \hat{z}_i = \beta_0 + \sum_j \beta_j \varphi_j(x_i)$$

X X β
 φ_j β
 X B-spline 4 16 RBF 140

5. B-spline

$$f(t) = \sum_k \beta_k B_k(t)$$

B-spline

20% 40% 60% 80% 4 1 120

6. sine/cosine

$$\sin(2\pi m/12) \quad \cos(2\pi m/12) \quad \sin(4\pi m/12) \quad \cos(4\pi m/12)$$

$m = 1 \dots 12$ B-spline
 12

7. RBF

RBF “ ” 1 0

$$\phi_k(s) = \exp(-||s-c_k||^2/(2\sigma^2))$$

140 16 $\sigma=0.8$ 16

RBF

8. Basis OLS

OLS log log

$$\beta_{OLS} = \operatorname{argmin}_{\beta} ||z-X\beta||^2$$

2014–2022 OLS 2023

Task 4

9. Ridge

B-spline RBF OLS Ridge

$$\beta_{Ridge} = \operatorname{argmin}_{\beta} \{||z-X\beta||^2 + \lambda ||\beta||_2^2\}$$

5 2018 2019 2020 2021 2022 λ 5 RMSE $\lambda=0.0132$ 2014
–2022 2023

Ridge L2 2022

10.

λ

$$MAE = \operatorname{mean}|y-\hat{y}| \quad RMSE = \sqrt{\operatorname{mean}((y-\hat{y})^2)} \quad R^2 = 1 - \frac{\sum(y-\hat{y})^2}{\sum(y-\bar{y})^2}$$

2018–2022 5 λ 2014–2022 2023 Global mean Neighbourhood mean
Basis OLS Basis Ridge

2023

	MAE	RMSE	R ²	
Global mean	2.170	4.164	-0.006	
Neighbourhood mean	1.391	2.699	0.577	
Basis OLS	1.120	1.966	0.776	RMSE R ²
Basis Ridge	1.054	1.976	0.774	MAE

Basis OLS 2023 RMSE 1.97 R² 0.776 Basis Ridge RMSE 1.98 MAE λ

- —
-
- $\log(1+y)$
- B-spline sine/cosine RBF
- Basis OLS Task 4 Basis Ridge L2
- 5 Ridge λ 2023

Part III •

“20” “ ”

1.

[illegible]

2. cross-validation K-fold

20 rolling-origin cross-validation

3. Task 4

20	Basis OLS	B-spline	RBF	Basis Ridge
	β		“ ”	

4. OLS Ridge

20						
OLS	Task 4	Ridge				
	RMSE	0.01	$\lambda=0.0132$	5	RMSE	2023

5. $R^2=0.774$ 77.4%

20

77.4%

R^2

MAE RMSE

6. $\log(1+y)$

20

log

7. 16 RBF

20

16

8.

20

Poisson negative-binomial

9.

downtown

20

reported theft counts

10.

Ridge

OLS

RMSE

20

RMSE

0.01

Ridge

OLS	RMSE	1.97	R ²	0.776	Ridge	RMSE	1.98	R ²	0.774	Ridge	MAE	λ
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- “ ”
- 5 2023
- OLS Ridge
- R^2